# Customer Shopping Behavior Analysis

*An End-to-End Data Analytics Project*

Python (Pandas) · SQL Server · Power BI

**Project Report**

# 1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior in order to guide strategic business decisions.

The analysis follows a complete data workflow: cleaning and preparing raw data in Python, loading it into a SQL Server database for structured querying, and building an interactive Power BI dashboard to communicate the findings visually.

## Tools & Technologies

| Stage | Tool / Technology |
|---|---|
| Data Cleaning & Preparation | Python (pandas, in Jupyter Notebook) |
| Database & Querying | Microsoft SQL Server (SSMS), SQLAlchemy + pyodbc |
| Visualization | Power BI |

# 2. Dataset Summary

- **Rows:** 3,900 customer transactions
- **Columns:** 18

**Key Features:**

  - Customer demographics: Age, Gender, Location, Subscription Status
  - Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color
  - Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

- **Missing Data:** 37 null values found in the Review Rating column

# 3. Data Preparation & Cleaning (Python)

The raw dataset was loaded and cleaned in a Jupyter Notebook before being pushed into SQL Server for analysis.

## 3.1 Data Loading & Initial Exploration

- Imported the dataset using pandas (pd.read_csv).
- Used df.info() to inspect structure and data types, and df.describe(include='all') to review summary statistics.

## 3.2 Missing Data Handling

- Identified 37 missing values, all in the Review Rating column, using df.isnull().sum().
- Imputed the missing ratings using the median rating within each product category, preserving category-level rating patterns rather than using a single global median.

### 3.3 Column Standardization

- Renamed all columns to snake_case (e.g. "Purchase Amount (USD)" → purchase_amount) for consistency and easier querying in SQL.

### 3.4 Feature Engineering

- **age_group:** customer ages were binned into four quartile-based segments — Young Adult, Adult, Middle-aged, and Senior — using pd.qcut().
- **purchase_frequency_days:** the text-based frequency_of_purchases field (e.g. "Weekly", "Quarterly") was mapped to an equivalent number of days, enabling numeric analysis of purchase cadence.

### 3.5 Data Consistency Check

- Compared discount_applied and promo_code_used and confirmed the two columns were identical across all records.
- Dropped the redundant promo_code_used column to avoid duplication.

### 3.6 Database Integration

The cleaned DataFrame was connected to a local Microsoft SQL Server instance (SAHIL-PC\SQLEXPRESS) using SQLAlchemy with the pyodbc driver, and loaded into a customer table in the customer_behavior database for SQL-based analysis:

```
from sqlalchemy import create_engine
from urllib.parse import quote_plus

server = r"SAHIL-PC\\SQLEXPRESS"
database = "customer_behavior"
driver = quote_plus("ODBC Driver 17 for SQL Server")

engine = create_engine(
    f"mssql+pyodbc://@{server}/{database}"
    f"?driver={driver}&trusted_connection=yes"
)

df.to_sql("customer", engine, if_exists="replace", index=False)
```

# 4. Data Analysis using SQL (Business Questions)

With the cleaned data loaded into SQL Server, ten business questions were answered using structured T-SQL queries.

### Q1. Total revenue generated by male vs. female customers

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender
```

| Gender | Revenue ($) |
|---|---|
| Female | 75,191 |
| Male | 157,890 |

### Q2. Customers who used a discount but still spent above the average purchase amount

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer)
```

Result: 839 customers met this criteria — high-spending customers who still respond to discount offers, an important segment for targeted promotions.

### Q3. Top 5 products by average review rating

```
SELECT TOP 5 item_purchased,
       ROUND(AVG(CAST(review_rating AS numeric)), 2) AS avg_rating
FROM customer
GROUP BY item_purchased
ORDER BY avg_rating DESC
```

| Item | Avg. Rating |
|---|---|
| Gloves | 3.86 |
| Sandals | 3.84 |
| Boots | 3.82 |
| Hat | 3.80 |
| Skirt | 3.78 |

### Q4. Average purchase amount: Standard vs. Express shipping

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2)
FROM customer
```

```
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type
```

| Shipping Type | Avg. Purchase ($) |
|---|---|
| Standard | 58.46 |
| Express | 60.48 |

## Q5. Do subscribed customers spend more?

```
SELECT subscription_status, COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount),2) AS avg_spend,
       ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
```

| Subscribed | Customers | Avg. Spend ($) | Total Revenue ($) |
|---|---|---|---|
| Yes | 1,053 | 59.49 | 62,645.00 |
| No | 2,847 | 59.87 | 170,436.00 |

Interestingly, average spend per customer is nearly identical between subscribers and non-subscribers — subscription status does not strongly influence individual spending, though non-subscribers drive far more total revenue simply due to volume.

## Q6. Top 5 products with the highest percentage of discounted purchases

```
SELECT TOP 5 item_purchased,
  ROUND(100.0 * SUM(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0 END)/COUNT(*), 2) AS
discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
```

| Item | Discount Rate (%) |
|---|---|
| Hat | 50.00 |
| Sneakers | 49.66 |
| Coat | 49.07 |
| Sweater | 48.17 |
| Pants | 47.37 |

## Q7. Customer segmentation — New, Returning, Loyal

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE WHEN previous_purchases = 1 THEN 'New'
         WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
         ELSE 'Loyal' END AS customer_segment
```

```
  FROM customer)
SELECT customer_segment, COUNT(*) AS num_customers
FROM customer_type GROUP BY customer_segment
```

| Segment | Number of Customers |
| --- | --- |
| Loyal | 3,116 |
| Returning | 701 |
| New | 83 |

The vast majority of the customer base (80%) falls into the Loyal segment, suggesting strong repeat-purchase behavior overall.

## Q8. Top 3 most purchased products within each category

```
WITH item_counts AS (
  SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer GROUP BY category, item_purchased)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts WHERE item_rank <= 3
```

| Rank | Category | Item | Total Orders |
|---|---|---|---|
| 1 | Accessories | Jewelry | 171 |
| 2 | Accessories | Sunglasses | 161 |
| 3 | Accessories | Belt | 161 |
| 1 | Clothing | Blouse | 171 |
| 2 | Clothing | Pants | 171 |
| 3 | Clothing | Shirt | 169 |
| 1 | Footwear | Sandals | 160 |
| 2 | Footwear | Shoes | 150 |
| 3 | Footwear | Sneakers | 145 |
| 1 | Outerwear | Jacket | 163 |
| 2 | Outerwear | Coat | 161 |

## Q9. Are repeat buyers (>5 previous purchases) more likely to subscribe?

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status
```

| Subscribed | Repeat Buyers |
|---|---|
| No | 2,518 |
| Yes | 958 |

Repeat buyers are more likely to not be subscribed (72%), indicating that subscription status is not a prerequisite for loyalty — a potential opportunity to convert already-loyal repeat buyers into subscribers.

## Q10. Revenue contribution by age group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC
```

| Age Group | Total Revenue ($) |
|---|---|
| Young Adult | 62,143 |

| Age Group | Total Revenue ($) |
|---|---|
| Middle-aged | 59,197 |
| Adult | 55,978 |
| Senior | 55,763 |

Revenue is fairly evenly distributed across age groups, with Young Adults contributing the most — but the spread (roughly 62K vs. 56K) shows no single age group dominates overall sales.

## 5. Dashboard in Power BI

The cleaned dataset was also connected directly to Power BI to build an interactive Customer Behavior Dashboard, allowing stakeholders to explore the same insights visually and filter by subscription status, gender, category, and shipping type.

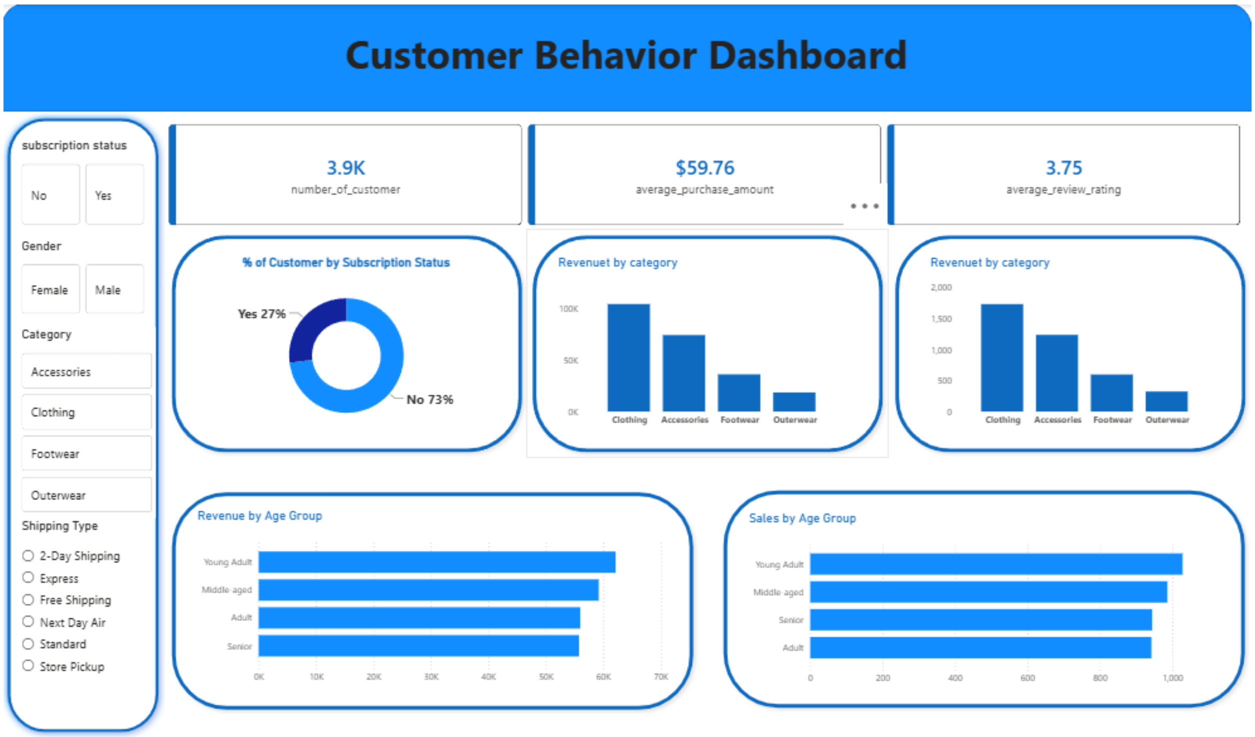


*Figure 1: Customer Behavior Dashboard (Power BI)*

**Dashboard components:**

- KPI cards: Number of Customers (3.9K), Average Purchase Amount ($59.76), and Average Review Rating (3.75).
- Donut chart: % of customers by subscription status (27% Yes / 73% No).
- Bar charts: Revenue by Category and Sales by Category.
- Bar charts: Revenue by Age Group and Sales by Age Group.
- Slicer panel: filters for Subscription Status, Gender, Category, and Shipping Type, enabling interactive drill-down.

## 6. Business Recommendations

- **Boost Subscriptions:** Promote exclusive benefits for subscribers, since only 27% of customers are currently subscribed despite subscribers showing comparable average spend to non-subscribers.

- **Customer Loyalty Programs:** Reward repeat buyers to move Returning customers into the Loyal segment and reinforce the behavior of the 3,116 customers already there.
- **Review Discount Policy:** Balance sales boosts with margin control — products like Hats and Sneakers are discounted on nearly half of all purchases.
- **Product Positioning:** Highlight top-rated products (Gloves, Sandals, Boots) and top-selling items per category (Jewelry, Blouse/Pants, Sandals, Jacket) in marketing campaigns.
- **Targeted Marketing:** Focus efforts on high-revenue age groups (Young Adults) and Express-shipping users, who show higher average purchase amounts.

## 7. Conclusion

This project demonstrates a complete analytics workflow — from raw data cleaning in Python, through structured business-question analysis in SQL Server, to an interactive Power BI dashboard. The findings highlight clear opportunities around subscription growth, loyalty conversion, and discount strategy that can directly inform business decisions.